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Final Report

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BSc Computer Science with Year Abroad

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**Beyond Step-Free: A
risk-based framework
for evaluating
accessible journey
reliability on the
London Underground**

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Abstract

Accessible journey planning for disabled passengers has seen significant improvements in recent years, with tools such as Google Maps and TfL Go offering step-free filtering and real-time accessibility information. However, these systems continue to optimise routes primarily based on travel time and represent accessibility as a binary property, failing to capture the reliability of accessible journeys and leaving disabled passengers exposed to routes that are technically classified as accessible but are in practice fragile, misleading, or unusable under real-world conditions.

A risk-based evaluation model is proposed and implemented in Python, assigning a numerical score to each route based on three measurable factors - interchange complexity, lift dependency, and destination step-free status. All routes are sourced directly from existing journey planning systems including Google Maps and TfL Go, ensuring the evaluation reflects genuine recommendations currently being made to disabled passengers. The model extends beyond static scoring to simulate disruption scenarios, revealing a category of vulnerability that existing systems do not capture: routes that appear maximally safe under normal conditions but become entirely infeasible under a single minor infrastructure fault.

The results demonstrate that what Google Maps classifies as a wheelchair accessible route can in practice score 9.6/10 on the risk scale, which is entirely unusable for a passenger requiring step-free exit. The sensitivity analysis confirms all twelve route classifications remain stable under $\pm 20\%$ weight variation, providing evidence of model robustness despite assumption-based penalty weights. The project concludes that accessibility risk can be modelled in a transparent, reproducible, and practically meaningful way, providing a foundation for future development of journey planning tools that better serve the disabled passengers who rely on these systems daily.

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1. Introduction

This chapter introduces the background, aim, and objectives of the project, focusing on the exploration of reliability within accessible journey planning for disabled users. It establishes the motivation for the research, presents the project's aim and objectives, and concludes with an overview of the report structure.

1.1 Background

Transport for London (TfL) provides a range of accessibility measures that intend to support disabled passengers, including step-free access information, door-to-door transport availability, accessibility maps and guides, as well as journey planning tools such as the TfL Go app, which allows users to filter their journey to step-free routes (Transport for London, 2026c). While these options show a significant improvement in tackling the on-going problem of representing disabled passengers fairly, users continue to experience high levels of journey disruption, followed by uncertainty and stress when using public transport.

About 15.7% of Londoners (around 1.2 million people) identify as disabled under the Equality Act 2010 (ONS, 2023), highlighting that this is a significant usability issue rather than a niche concern. Survey evidence shows that roughly 44% of disabled passengers have experienced problems with lifts on their recent journeys, including outages, lifts being too small for larger wheelchairs or mobility scooters, and too few lifts to meet demand (Transport for All, 2023). Furthermore, reporting based on TfL operational data indicates that step-free access suspensions due to

staffing issues were on course to reach 1,200 in 2022-23, up from just 20 in 2018-19 (Disability News Service, 2022).

These recurring issues leave us thinking of an appropriate solution to this problem. The TfL Go app provides real-time accessibility information, including lift availability, most commonly in their app and online services, however this is often presented separately from journey planning and does not convey the overall fragility and reliability of an accessible journey. A journey planned using another option, Google Maps similarly provides some accessibility-related information but also requires users to manually interpret station-level details and assess the risk of available routes. This project aims to explore the highlighted gap in existing journey planning approaches, the starting point being evaluating accessible routes in terms of reliability primarily, and not travel time.

1.2 Aim

The aim of this project is to conduct a thorough investigation into the reliability of accessible journey planning within public transport networks, as well as designing and implementing a risk-based evaluation model which goes beyond the limitations of existing systems. I am going to ground this project in the context of Transport for London's network, focusing on real-world journey scenarios between a range of origin and destination pairs. This provides a concrete and well-documented foundation for the research, given the availability of TfL accessibility data and the relevance of the problem to a great portion of the population of disabled users in London.

The model I will develop in this project takes as input a set of possible routes and evaluates each one not primarily on travel time, but on its susceptibility to disruption - considering factors such as interchange complexity, lift dependency, and step-free access availability at the destination. This allows the system to distinguish between routes that are technically accessible and those that are genuinely reliable under real-world conditions, addressing the gap that will be identified later, in Chapter 2.

The ideal outcome of this project is a working, reproducible risk scoring system that can meaningfully differentiate between accessible routes and provide a more honest representation of journey reliability than what current tools offer. If I manage to deliver this, I will demonstrate that accessibility-related risk factors can be modelled in a transparent and practical way, providing a foundation that could influence future development of journey planning tools used by disabled passengers every day.

1.3 Objectives

This subchapter outlines the objectives pivotal while creating my project, focusing on the identification of reliability-related risk factors, the design of an evaluation model, and finishing with a critical assessment of its outputs.

1. Carry out a thorough literature review of existing journey planning systems and accessibility research to gain an understanding of what we're working with
 - 1.1 Review existing journey planning applications to identify how accessibility is currently represented and what is going well
 - 1.2 Identify the key limitations of existing systems, especially in relation to reliability and infrastructure dependency
2. Define and justify the risk factors to be incorporated into the evaluation model:
 - 2.1 Establish which factors most significantly affect the reliability of accessible journeys, drawing on TfL operational data and academic literature
 - 2.2 Figure out a weighting approach that reflects the relative severity of each factor
3. Design and implement a risk scoring system in a programming tool, such as Python, that is capable of evaluating accessible routes:
 - 3.1 Represent routes and their associated accessibility properties as structured data

- 3.2 Implement a function that works out a (normalised) risk score for each route
- 3.3 Simulate disruptions to assess how route reliability changes under infrastructure failure scenarios
- 4. Evaluate the model against real journey scenarios on the tube:
 - 4.1 Apply the model to a set of real journey scenarios on the London Underground network, each with multiple, varying route options
 - 4.2 Compare routes based on their risk scores, and then travel time to assess the trade-offs involved
 - 4.3 Export route evaluation results to a CSV file and apply a data visualisation tool to illustrate my key findings and patterns across routes, as well as generate insights

1.4 Report Structure

I've decided to structure this report into six chapters. Chapter 1 introduces the background, motivation, and scope of the project, establishing the aim, objectives, and research question that will help me guide my work. In Chapter 2, I will present a comprehensive literature review covering accessibility in public transport, existing journey planning applications, and current approaches to route optimisation. It identifies the key limitations of existing systems and concludes by establishing the research gap that this project aims to address. Chapter 3 will describe the design and methodology of the proposed risk solution, detailing the risk factors considered, the scoring system developed, and the assumptions and design decisions made throughout the process. Chapter 4 will cover the implementation of the system, including data visualisation of the results. Chapter 5 will present the evaluation of the model across a set of real journey scenarios, comparing routes on the basis of both travel time and risk score and discussing the findings in relation to the research question. At last, Chapter 6 will conclude the report with a summary of findings, a critical discussion of the model's limitations, directions for future work, and a final conclusion.

1.5 Research Question

The following research question guides the investigation carried out in this project:

How can accessibility-related risk factors be modelled and integrated into journey planning systems in order to improve the reliability and usability of accessible routes for disabled users?

This question is addressed through the design and implementation of a risk-based evaluation model, and is evaluated against real journey scenarios on the London Underground network. The findings are presented in Chapter 5 and critically discussed in Chapter 6.

2: Literature Review

2.1 Introduction

This section is a comprehensive literature review on accessible journey planning within public transport systems, with a focus on step-free access and how accessibility is represented in existing journey planning applications. It examines current approaches to route optimisation and the limitations of widely used systems such as Google Maps, TfL Go, and Citymapper. The chapter finishes by identifying a gap in the consideration of journey reliability, which forms the foundation for the approach proposed in this project.

2.2 Existing Journey Planning Systems

As I mentioned previously in Chapter 1, disabled passengers in London face significant and frequent disruption when using public transport. The digital tools that are designed to support journey planning for these passengers have improved considerably in recent years, and represent a meaningful opportunity to improve independence and accessibility for disabled users.

Transport for London's official journey planning application, TfL Go, is arguably the most detailed accessibility-focused tool available to London passengers. The app includes a step-free mode filter which can be activated by tapping on the step-free button to update the map to only show step-free stations (Transport for London, 2026c). Passengers can also access dedicated guidance on wheelchair access and avoiding stairs across the network directly through TfL's accessibility pages (Transport for London, 2025a). Citymapper, another widely used third-party alternative, offers step-free routing, with routes that prioritise simplicity over travel times and walking times adjusted for people with reduced mobility, avoiding stairs, wide gaps, and complicated transfers (Citymapper, 2023). Google Maps, similarly to TfL Go, also provides a 'Wheelchair accessible' filter, which is designed to direct users toward step-free entrances and exits.

Despite this potential, research consistently shows how exactly existing systems fall short of meeting the needs of disabled passengers in practice. A survey of over 1,200 disabled people conducted by the National Centre for Accessible Transport found that 62% use websites and 57% use apps for journey planning, but also that 30% still rely on friends, family, or carers to plan their journeys, suggesting that current digital tools are not sufficiently independent or comprehensive to meet all of their needs (NCAT, 2025). The same study found that although 66% of respondents trusted Google Maps most for journey planning, 31% of those same users found its accessibility data to be inaccessible or incorrect - highlighting a significant gap between the availability of these tools and the quality of the information they provide. Additionally, 23% of respondents reported that accessibility information when planning journeys is limited, and 21% noted that information on vehicles is unavailable or inaccessible (NCAT, 2025). Transport operators providing open data do not require accessibility compliance from the third-party developers who use it, meaning the quality of accessibility information across different journey planning tools depends entirely on whether individual developers choose to prioritise it, as no regulatory obligation exists to ensure they do (Gebresselassie and Baljko, 2025). As highlighted by Transport for All, limited research exists on whether these planning tools genuinely make journeys easier for disabled passengers, (Transport for All, 2023) and the shortage of research I

have personally encountered demonstrating positive experiences of disabled users with journey planning apps, from the present, or past, correlates further with the goal of this project.

My findings suggest that while accessibility has now become a serious consideration within journey planning, its implementation remains inconsistent and not reliable enough for disabled users. I examine the specific ways in which these systems fall short in greater detail in section 2.5.

2.3 Accessibility in Public Transport

Accessibility in public transport can be defined as the ability of individuals to reach desired destinations and opportunities using available transport systems (Geurs and van Wee, 2004). It extends beyond physical infrastructure to include usability, service availability, and the ability for individuals to travel safely and independently. A key component of accessibility within urban transport networks is step-free access, which enables movement between street level, platforms, and vehicles without the use of stairs.

Within real-world transport systems however, accessibility is often limited and inconsistent. Within the London Underground network, step-free access remains a flaw despite ongoing improvements in recent years. Transport for London reports that at this moment, 94 out of 272 Tube stations are step-free, meaning that approximately one third of the network provides step-free access (Transport for London, 2026a). Although this represents a significant improvement to what the number was 3, 4 years ago, accessibility across the network remains uneven, meaning that many journeys still depend on specific stations, interchanges, or alternative routes to remain fully accessible.

TfL is continuing to expand step-free access across the network, but factors such as the reliability of accessibility features, the complexity of station layouts, and environmental conditions remain critical, and may become increasingly important in the near future as more journeys will depend on accessibility infrastructure. For example, lift availability plays a central role in enabling step-free travel, yet is frequently subject to disruption. Transport for London data reports that there were

7,598 lift and escalator faults recorded across the network in 2025 (Transport for London, 2026b).

Station	OOS Fault Count	Average of Repair time (hrs)
Waterloo	451	35:40
Tottenham Court Road	304	9:39
King's Cross St. Pancras	298	18:13
Victoria	277	26:22
Liverpool Street	251	2:45
London Bridge	251	9:53
Bank	235	23:25
Stratford	220	20:04
Canary Wharf (Elizabeth line)	215	13:00
Green Park-	195	6:37

Total Lift and Escalator OOS faults	7598	
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*OOS – Out of service

Figure 1: Top 10 London Underground stations by lift and escalator out-of-service faults (Transport for London, 2026b)

The stations shown in Figure 1 are all at least partially step-free, while simultaneously being some of the most complex and heavily used interchanges in the whole network, including King’s Cross St. Pancras, Waterloo, or Victoria. The table makes it evident that more complex stations may introduce additional points of failure, and reinforces the idea that the presence of accessibility infrastructure alone does not guarantee reliability. It is therefore crucial to understand that accessibility in public transport is a complex and context-dependent concept that can vary significantly between routes.

2.4 Journey Planning and Optimisation Approaches

Journey planning applications rely on optimisation techniques to identify routes based on criteria such as travel time, walking distance, and the number of

interchanges. The primary objective, in most cases, is to minimise total journey duration, with alternative routes commonly presented as secondary options. Van Oort and Cats (2015) emphasise the importance of incorporating data-driven approaches to improve planning and operational decision-making, particularly in relation to system performance and reliability, suggesting that effective route evaluation should consider not only efficiency, but also the variability and robustness of transport services.

More adaptive approaches to route planning have been explored as of recent, including the use of large language models (LLMs) to enable more context-aware decision-making (Jonnala et al., 2024), as well as methods that explicitly account for dynamic conditions such as infrastructure unavailability (Prakash, Mehta and Sharma, 2024). The fact that such developments are needed highlights how limited existing systems remain when faced with the complexity of real-world travel conditions, and push a shift towards more adaptive and intelligent route planning methods.

Another development in this space is the move towards personalised, user-specific routing. Salanova Grau et al. (2025) discuss a multimodal trip planner that tailors route recommendations based on individual user preferences, captured through historical travel behaviour. This shows the growing capability of journey planning systems to deliver tailored recommendations and not just presenting all users with identical route options. Despite this, personalisation in existing systems has not been extended to account for the reliability of accessible infrastructure - meaning that even as systems become more individualised, disabled users are still not receiving routes that reflect the real-world fragility of the journeys available to them.

Despite these advances, most widely used journey planning applications continue to prioritise travel time as the dominant factor in route selection. Accessibility is generally incorporated as a constraint, rather than as a factor influencing route quality. This limits the ability of existing systems to differentiate between routes that may vary significantly in terms of reliability, leading directly to the limitations discussed in the following section.

2.5 Limitations of Current Systems

Even though modern journey systems provide a range of accessibility related features, multiple limitations remain in how accessible routes are evaluated and then presented to the user. The gaps are not specific to a singular application, but are rather observed across the most popular systems that are used every day by millions of people. The following subsections outline the key issues identified.

2.5.1 Prioritisation of travel time over journey reliability

Existing journey planning systems primarily optimise routes based on travel time, even when accessibility filters are applied. Google Maps and Citymapper both output the journeys as a sorted array from the fastest available route down, while TfL Go similarly presents routes ordered by estimated journey duration, providing additional labeling such as 'Arrives first', 'Alternative', or 'Bus only' to distinguish between options.

Although the TfL Go labels offer partial guidance, they do not change the underlying idea of prioritising routes based on shortest travel time. As a result, users may still be influenced to take faster routes that depend on complex interchanges or accessibility infrastructure, increasing the likelihood of disruption. This highlights a limitation in current systems, as reliability is not explicitly considered alongside travel time in route ranking. The prioritisation contributes to broader issues in how accessibility is represented within journey planning systems.

2.5.2 Binary representation of accessibility

Accessibility within public transport systems is a complex and multi-dimensional concept, influenced by infrastructure, environmental conditions, and individual user needs. However, research in transport accessibility shows that for evaluation purposes, accessibility is often converted into measurable indicators, which simplifies real-world conditions (Geurs and van Wee, 2004).

In digital journey planning systems, this simplification results in outputs that effectively present accessibility as a binary decision from the user's perspective,

where routes are labelled as accessible or not accessible based on predefined criteria. Although these systems may rely on more detailed data internally, that is not what is fully reflected in the user interface. As a result, distinctions between full step-free access and partial accessibility, such as stations that only support interchanges between platforms, are not clearly communicated. The simplified representation of accessibility can lead to misleading outputs in practice, as demonstrated in the following example.

2.5.3 Misleading outputs

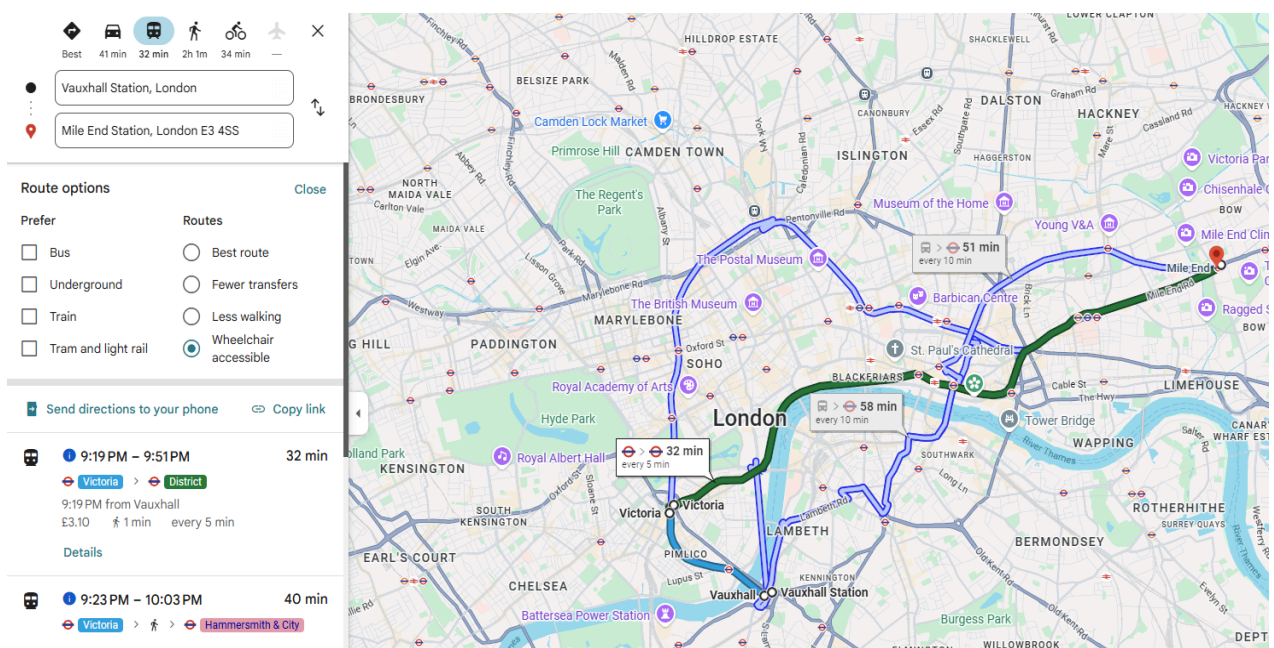


Figure 2: Google Maps identifying a route as wheelchair accessible despite incomplete step-free access at Mile End station.

As shown in Figure 2, Google Maps classifies the route as wheelchair accessible, despite the destination lacking full step-free access to street level. Mile End station is only partially accessible, supporting interchange platform-to-platform, but none from platform to street.

This mistake highlights a limitation in how accessibility is represented, as the system does not distinguish between partial and complete accessibility. As a result, users may be presented with journeys that are technically feasible within the system's definition, but not usable in real-world scenarios.

While this example is captured from Google Maps, similar limitations can be observed across the whole domain, where accessibility information is presented at a high level without fully capturing station-level constraints. This strengthens the need for a more detailed and reliability-focused alternative in evaluating accessible routes.

2.5.4 Lack of reliability modelling

Existing journey planning systems provide accessibility-related information. However, they do not explicitly model the reliability of accessible routes. Paths that are satisfying step-free access constraints are treated as equally viable, regardless of differences in their needs to depend on infrastructure, such as lifts, or how complex an interchange is. In the real world, accessible journeys can vary significantly in terms of reliability, with routes that require several interchanges or rely heavily on lift infrastructure being more susceptible to breakdowns, causing disruption.

This limitation is very important considering that route feasibility in real-world transport systems is inherently dynamic. Recent research highlights the need to account for disruptions and infrastructure unavailability when evaluating routes, as these factors can directly affect usability (Prakash, Mehta and Sharma, 2024). This suggests that accessibility should not be treated as a fixed property of a route, but rather as something that can be changed depending on system conditions.

2.6 Conclusion and identified research gaps

This chapter has reviewed existing journey planning systems, the concept of accessibility in public transport, and modern approaches to route optimisation. While current applications provide accessibility-related features, they continue to prioritise travel time and rely on simplified representations of accessibility. The review has demonstrated that these systems do not adequately reflect the real-world conditions that affect accessible travel, particularly in terms of reliability and infrastructure dependency. As a result, important factors influencing the usability of accessible routes are not fully captured within existing journey planning approaches.

This highlights a key gap: current journey planning applications do not account for the reliability of accessible routes or their susceptibility to disruption. My project aims to address this gap by developing a reliability-focused approach to accessible journey planning, which evaluates routes not only based on accessibility constraints, but also on their robustness in real-world conditions. Circling back to the research question outlined earlier in this report, the solution addressing the research gap will aim to provide a more realistic and practical framework for evaluating accessible journeys, incorporating both accessibility and reliability as core factors in route selection.

3: System Design / Methodology

3.1 Introduction

This chapter introduces the design and methodology of the risk-based evaluation model. It begins with an overview of the proposed approach, before defining the three main risk factors incorporated into the model, each justified with reference to the limitations identified in Chapter 2. The risk scoring system is then described, including raw score calculation, normalisation, and classification bands. The chapter concludes with a discussion of key assumptions and design decisions, and what these mean for the model's scope and generalisability.

3.2 Overview of Proposed Approach

This project's goal is to develop a reliability-focused journey planning model for accessible travel within public transport networks. Unlike existing journey planning applications, which primarily optimise routes based on travel time, the proposed system evaluates routes based on their reliability, particularly in the context of step-free accessibility. This shift reflects the need to consider not only whether a route is technically accessible, but also whether it remains usable under real-world conditions, where service variability and disruptions can significantly affect transport performance (van Oort and Cats, 2015).

The system takes as input a set of possible routes between a given origin and destination. Each route is represented as a sequence of stations and

interchanges, alongside associated accessibility information such as step-free availability and dependence on infrastructure including lifts. This representation allows the system to analyse routes at both a global level (the overall journey) and a local level (individual stations and transitions between them).

For each route, the system evaluates a set of factors that influence its reliability, which are derived from the limitations identified in Chapter 2 and intended to capture the primary sources of disruption affecting accessible journeys. Particularly, the system considers the number of interchanges required, the degree of reliance on accessibility features, and the structural complexity of stations involved in the route. Each of these elements represents a potential point of failure, where the usability of a route may be compromised, particularly under conditions of infrastructure unavailability or operational disruption (Prakash, Mehta and Sharma, 2024).

The system for this project then evaluates these factors to determine how susceptible a route is to disruption. It does not treat accessibility as a binary condition like previously discussed existing systems, allowing the model to distinguish between routes that are technically accessible but fragile, and those that are more robust under real-world conditions. Based on this evaluation, the system computes a reliability score for each route, reflecting the likelihood that it will remain usable in practice. Routes are then ranked according to this score, enabling comparison not only in terms of efficiency, but also in terms of practical usability. The model also supports the simulation of disruption scenarios, to be precise minor and major lift failures at specific stations, allowing route reliability to be assessed under changing conditions.

At last, results are output in a structured format, enabling me to carry out further analysis and visualisation, as well as a comparison between routes and identification of patterns in reliability.

3.3 Risk Model Design

The risk model proposed in this project is designed as a scoring system that evaluates the reliability of an accessible route by quantifying the key factors

identified in Chapter 2. It assigns a numerical risk score to each route rather than treating accessibility as a binary property, showing its overall susceptibility to disruption. A higher score makes a route more fragile under real-world conditions, whereas a lower score makes one that is more robust and dependable.

The model operates on a raw scoring basis, where individual risk factors contribute additive penalty values to a cumulative total. Each penalty is calibrated to reflect the relative significance of the corresponding factor in terms of its potential impact on journey usability. For instance, a route that ends at a station without step-free access to street level represents an absolute barrier to journey completion for many disabled users, and is therefore assigned a substantially higher penalty than a route with a moderately complex interchange. This weighting approach ensures that the most critical failure points have a greater influence on the overall score.

Once all applicable penalties have been applied, the raw score is normalised to a scale of 0 to 10 using a defined maximum value, allowing scores to be interpreted consistently across different routes and journeys. This normalised score is then mapped to one of three classification bands: low, medium, or high risk, which provides a clear and actionable output for route comparison.

This weighted additive structure is consistent with the principles of Multi-Criteria Decision Analysis (MCDA), a family of methods that aggregate multiple competing criteria into a single comparable score to support decision-making. The model applies this principle to route evaluation, treating interchange complexity, lift dependency, and destination accessibility as distinct criteria that together determine overall route reliability. Unlike formal MCDA approaches, which typically derive weights through structured stakeholder elicitation processes such as surveys or expert panels, the weights in this model are defined through logical reasoning grounded in the literature reviewed in Chapter 2. This was a deliberate scoping decision - formal elicitation would require access to disabled passengers and domain experts beyond the resources available for this project, and the primary contribution here is the framework itself rather than a statistically optimised parameterisation of it. The sensitivity analysis in section 5.4 provides

evidence that the classifications produced are stable under reasonable weight variation, partially mitigating this limitation, though empirical derivation remains the most important direction for future work as discussed in Chapter 6.

The model is intentionally kept transparent and rule-based rather than probabilistic. This design decision reflects my scope of the project, which prioritises interpretability and reproducibility over statistical complexity. A rule-based approach also makes the model's reasoning auditable, because each penalty can be traced back to a specific route property, making it straightforward to explain why one route scores higher than another. This is important in the context of accessibility, because passengers may need to understand and trust the rationale behind what my model recommends.

It is acknowledged that this approach involves simplifying assumptions. The penalty values are defined based on logical reasoning with close attention to the literature reviewed in Chapter 2, rather than empirically derived from historical disruption data. The implications of this are discussed further in Chapter 6.

3.4 Definition of Risk Factors

This section defines the three core risk factors incorporated into the model, each representing a distinct source of potential disruption to an accessible journey. The factors were selected based on the limitations identified in Chapter 2, particularly the evidence that infrastructure dependency and interchange complexity are primary contributors to journey unreliability for disabled passengers (Prakash, Mehta and Sharma, 2024).

3.4.1 Interchange complexity

An interchange is a point within a journey where a passenger has to transfer between lines or services. For disabled passengers, these represent a heightened point of risk, as they typically require navigating between platforms using lifts or other accessibility infrastructure. The model accounts for interchange complexity at three levels, which are low, medium and high - reflecting the structural difficulty of navigating a given station. High complexity stations involve a greater number of platforms, longer transfer distances, and higher lift dependency. As demonstrated

in Chapter 2, the stations with the highest recorded lift faults are also the most structurally complex interchanges on the network (Transport for London, 2026b), reinforcing the relationship between complexity and reliability risk. The number of interchanges required also contributes a base penalty independent of complexity, reflecting the cumulative risk introduced by each additional transfer point.

3.4.2 Lift dependency

Lift dependency refers to the total number of lifts required to complete a journey, with each lift representing an individual point of failure. As discussed in Chapter 2, the frequency of lift faults on the London Underground network is well-documented, and survey evidence confirms that a significant proportion of disabled passengers have experienced disruption as a direct result (NCAT, 2025). The model assigns penalties on a tiered basis depending on the total lift count across all interchanges - low, moderate, and high, reflecting the increased probability that at least one lift will be unavailable at the time of travel as the total number rises.

3.4.3 Accessibility constraints

Accessibility constraints refer to the step-free status of the journey's origin and destination. If the origin station is not step-free, the journey is classified as entirely infeasible, as the passenger would be unable to begin travelling. If the destination lacks full step-free access to street level, a substantial penalty is applied - reflecting my finding discussed in section 2.5.3 that reaching a destination platform does not guarantee the ability to exit the station independently. This directly addresses the gap in existing systems where partial accessibility is misrepresented as full accessibility.

3.4.4 Station complexity classification

The classification of interchange stations into low, medium, and high complexity tiers is a main component of the risk model, as it influences the penalty applied to each route. In this section I define the criteria used to assign each classification and provide visual evidence to support the distinctions made.

I have evaluated each station against three criteria: the number of lifts required to navigate the interchange, the number of distinct platform levels involved in the transfer, and the station's recorded annual passenger volume. Stations scoring high across all three criteria are classified as high complexity, those with moderate characteristics are classified as medium, and those with minimal requirements across all criteria are classified as low. Where it is applicable, recorded fault frequency from Transport for London's lift and escalator performance data (Transport for London, 2026b) is used as a further indicator, because of the strong correlation observed in Chapter 2 between structural complexity and fault frequency. I've also decided to factor in the annual passenger count based on the influence it has on disruptions. All other interchange stations in the model were classified using these same criteria, with the three examples below serving as the definitive reference points for each tier.

The following table summarises one representative station per classification tier:

Station	Classification	Lifts required	Platform levels	Annual count (by journeys, 2025)	Lift faults (2025)
Bank / Monument Station	High	4	Multiple	High (40.527 million)	41
Green Park Station	Medium	2	3	Medium (29.186 million)	29
Westferry Station	Low	2	1	Low (4.193 million)	—

Table 1: Representative station complexity classifications, showing lifts required, platform levels, annual passengers and lift faults per tier. Annual passenger figures sourced from Transport for London (2025b). Lift fault figures sourced from a Transport for London FOI request (2025c). Westferry does not appear in the TfL lift fault dataset, consistent with its low infrastructure dependency.

The visual contrast between complexity tiers is illustrated by the station layout diagrams in Figures 3, 4 and 5.

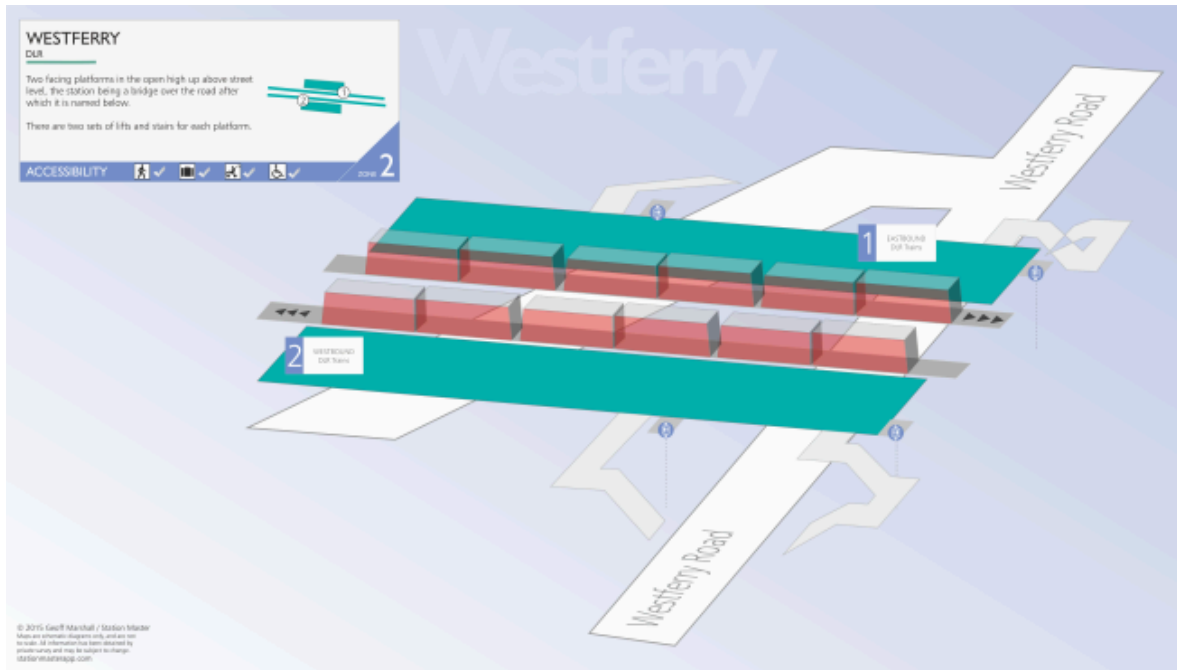


Figure 3: Station layout diagram of Westferry DLR station (Station Master, 2015), a low complexity interchange with two platforms above street level and one lift per platform.

Figure 3 shows Westferry DLR station, the low complexity reference point for this model. The layout consists of two facing platforms situated above street level, with two sets of lifts - one per platform. The structure is compact and straightforward, presenting minimal navigational challenge for a disabled passenger and limited infrastructure dependency.

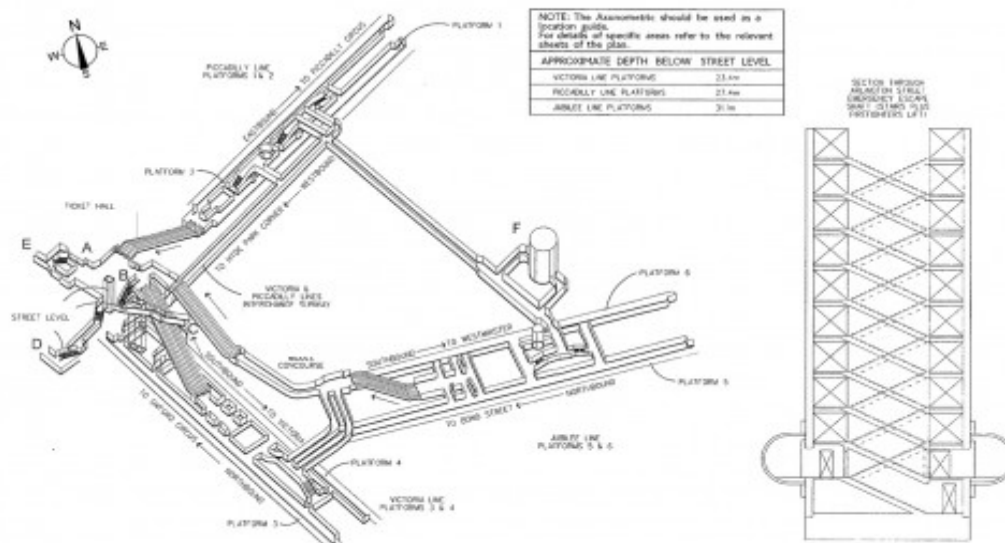


Figure 4: Axonometric layout of Green Park station (Transport for London, 2015), a medium complexity interchange

Figure 4 shows Green Park station, the medium complexity reference point. The diagram reveals three line platforms, precisely Victoria, Piccadilly, and Jubilee - at three distinct depths below street level: 23.4m, 27.4m, and 31.1m respectively (Transport for London, 2015). Navigating between lines requires both vertical and horizontal movement via the interchange subway visible in the diagram, with lifts required at each level transition. Green Park recorded 29 lift faults in 2025, consistent with the medium infrastructure dependency claim (Transport for London, 2025c).

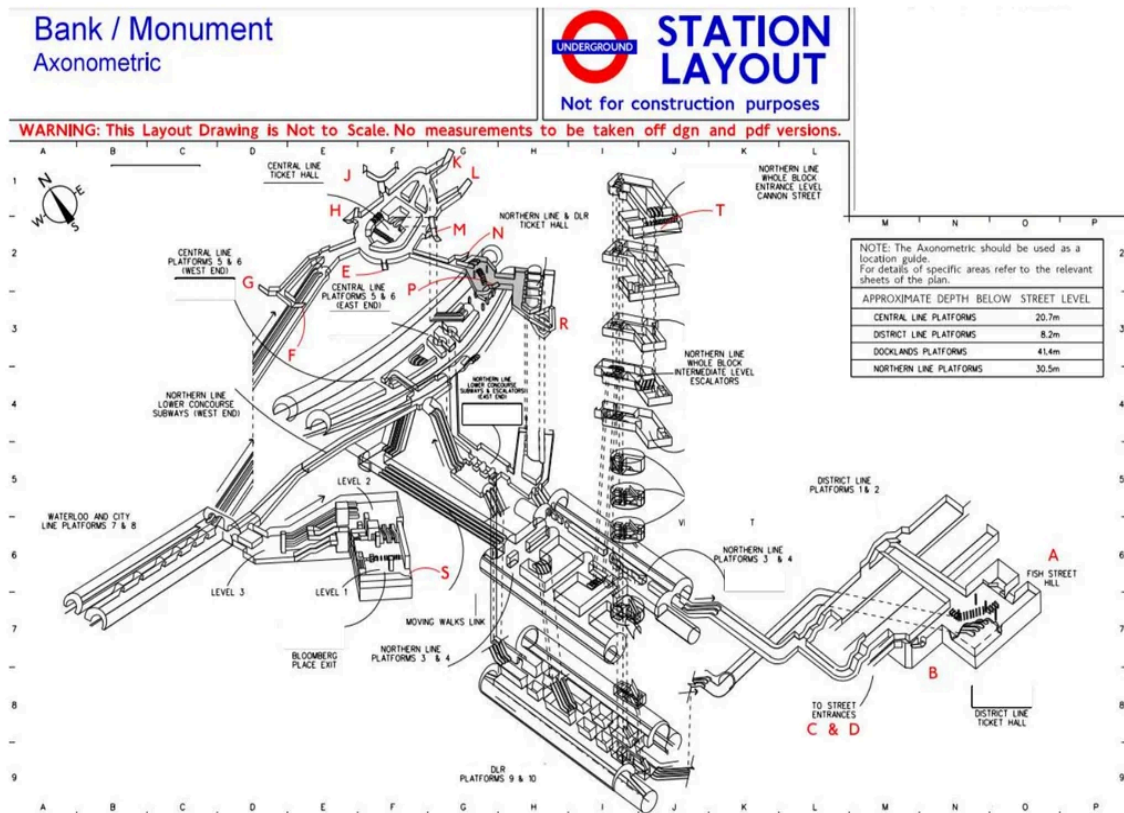


Figure 5: Axonometric layout of Bank/Monument station (Transport for London, 2023), a high complexity interchange with platforms across multiple levels for five lines connected by an extensive network.

Figure 5 shows Bank/Monument Station, the high complexity reference point. The diagram illustrates the scale and structural intricacy of the station, with the Central, Northern, Waterloo & City, DLR and District line platforms spread across multiple levels, connected by an extensive network of subways, escalators and lifts. Navigating between the Northern line and DLR platforms, as required in some journeys examined later in this project, involves multiple level changes and significant horizontal distance. Bank recorded 41 lift faults in 2025, which further supports its high complexity classification (Transport for London, 2025c).

Taken together, I believe these diagrams demonstrate that the three-tier classification system reflects genuine and observable differences in station

structure, and that the penalty values assigned to each tier are proportionate to the navigational demands placed on disabled passengers at each interchange.

3.5 Risk Scoring System

This section will go into detail on how the risk factors defined in 3.4 are translated into a numerical score for each route. The scoring system operates in three stages: raw score calculation, normalisation, and classification.

3.5.1 Raw score calculation

The overall risk score for a route r is computed as a weighted sum of its constituent risk factors, expressed as:

$$R(r) = I(r) + C(r) + L(r) + A(r)$$

where:

- $R(r)$ is the total raw risk score for route r
- $I(r)$ is the interchange count penalty, defined as the number of interchanges multiplied by a base penalty of 1
- $C(r)$ is the interchange complexity penalty, defined as the sum of complexity penalties across all interchanges, where low complexity contributes 1 point, medium complexity contributes 2 points, and high complexity contributes 4 points
- $L(r)$ is the lift dependency penalty, assigned on a tiered basis according to the total number of lifts required across all interchanges
- $A(r)$ is the accessibility constraint penalty, applied based on the step-free status of the destination station

Each route is evaluated by applying additive penalties corresponding to these factors. The penalties are cumulative, meaning a route that scores poorly across multiple factors receives a proportionally higher raw score than one affected by a single factor. The specific penalties applied are as follows:

- Each interchange contributes a base penalty of 1 point regardless of complexity, reflecting the cumulative risk introduced by every additional transfer point
- A medium complexity interchange contributes an additional 2 points, while a high complexity interchange contributes 4 points
- Lift dependency is assessed on a tiered basis, where a total of 1 lift contributes 1 point, 2 or more lifts contributes 3 points, and 5 or more lifts contributes 5 points, reflecting the increased probability of at least one failure as dependency grows
- If the destination station does not have full step-free access to street level, a penalty of 14 points is applied, reflecting the severity of this constraint as a near-absolute barrier to journey completion
- If the origin station is not step-free, the journey is classified as infeasible and assigned a maximum raw score of 25, removing it from further comparison

3.5.2 Normalisation

Once the raw score has been calculated, it is normalised to a scale of 0 to 10 using the following formula:

$$N(r) = \min((R(r) / 25) \times 10, 10)$$

A maximum raw score of 25 was defined to represent the worst realistic case of a feasible journey - one that involves multiple high complexity interchanges, high lift dependency, and a destination that is not fully step-free. Normalisation ensures that scores are interpretable and comparable across different journeys and routes, regardless of the number of risk factors present.

3.5.3 Classification

The normalised score is mapped to one of three classification bands, providing a clear and actionable summary of route reliability:

- A score of 0 to 3.9 is classified as 'Low' risk, indicating a route that is likely to remain usable under most real-world conditions

- A score of 4.0 to 6.9 is classified as ‘Medium’ risk, indicating a route with meaningful infrastructure dependency that may be affected by disruption
- A score of 7.0 or above is classified as ‘High’ risk, indicating a route that is fragile under real-world conditions and likely to be affected by even minor disruption

These thresholds were defined based on the distribution of scores produced by the model across the journeys evaluated in Chapter 5, ensuring that the classification bands produce a meaningful and informative spread of results.

3.5.4 Disruption severity

The model extends beyond static scoring by simulating disruption scenarios at specific interchange stations. Two severity levels are defined, grounded in the repair time data presented in Figure 1 of Chapter 2 (Transport for London, 2026b):

A minor disruption simulates a short-duration fault where one lift at the affected station is temporarily out of service. For stations with multiple lifts, this reduces the lift count by one, potentially pushing the route into a higher lift dependency penalty tier. For stations with a single lift, even a minor fault renders the interchange impassable, immediately classifying the route as infeasible. This reflects the real-world scenario where stations with no lift redundancy offer no fallback when infrastructure fails.

On the other hand, a major disruption simulates an extended outage where the interchange is rendered entirely impassable regardless of its lift count, reflecting faults with repair times exceeding 20 hours as recorded in the TfL data. Under a major disruption, the affected route is classified as infeasible irrespective of its base risk score.

This distinction allows the model to differentiate between disruption scenarios of varying severity and produces findings that would not be visible from base risk scores alone. In particular, it surfaces a class of routes that appear low risk under normal conditions but collapse entirely under even minor disruption, which is a vulnerability that existing journey planning systems do not capture.

3.6 Assumptions and Design Decisions

The risk model was developed with some simplifying assumptions that define the boundaries within which its findings should be interpreted. Routes are manually defined rather than algorithmically generated, which was a deliberate decision of mine to focus on the reliability scoring model as the novel contribution of this project. Penalty weights are influenced from the literature reviewed in Chapter 2 rather than empirically derived from historical disruption data, and interchange complexity classifications are assigned based on station layout documentation and TfL operational data rather than a formally computed metric. The model produces static scores rather than dynamically updating in response to real-world conditions, and the disruption severity levels represent a simplification of what is in practice a continuously varying phenomenon. Finally, the model's scope is limited to the London Underground network and selected bus routes, however a similar approach can be implemented in any other metro network in the world. The implications of these decisions and directions for addressing them are discussed in Chapter 6.

4: Implementation

4.1 Architecture of the System

The system is implemented in Python as a single-file script, reflecting my emphasis on the project being transparent and reproducible. The architecture follows a straightforward pipeline - route data is defined as structured dictionaries, passed through a risk calculation function, normalised and classified, then evaluated across disruption and sensitivity scenarios before results are exported to CSV files for visualisation.

The system is composed of six functional components: route data definitions, core risk logic, journey evaluation, disruption simulation, sensitivity analysis, and CSV export. Each component is modular and independently testable, meaning individual functions can be called in isolation to verify their behaviour. This design

also makes it straightforward to extend the system with additional routes, journeys, or penalty factors without modifying existing logic.

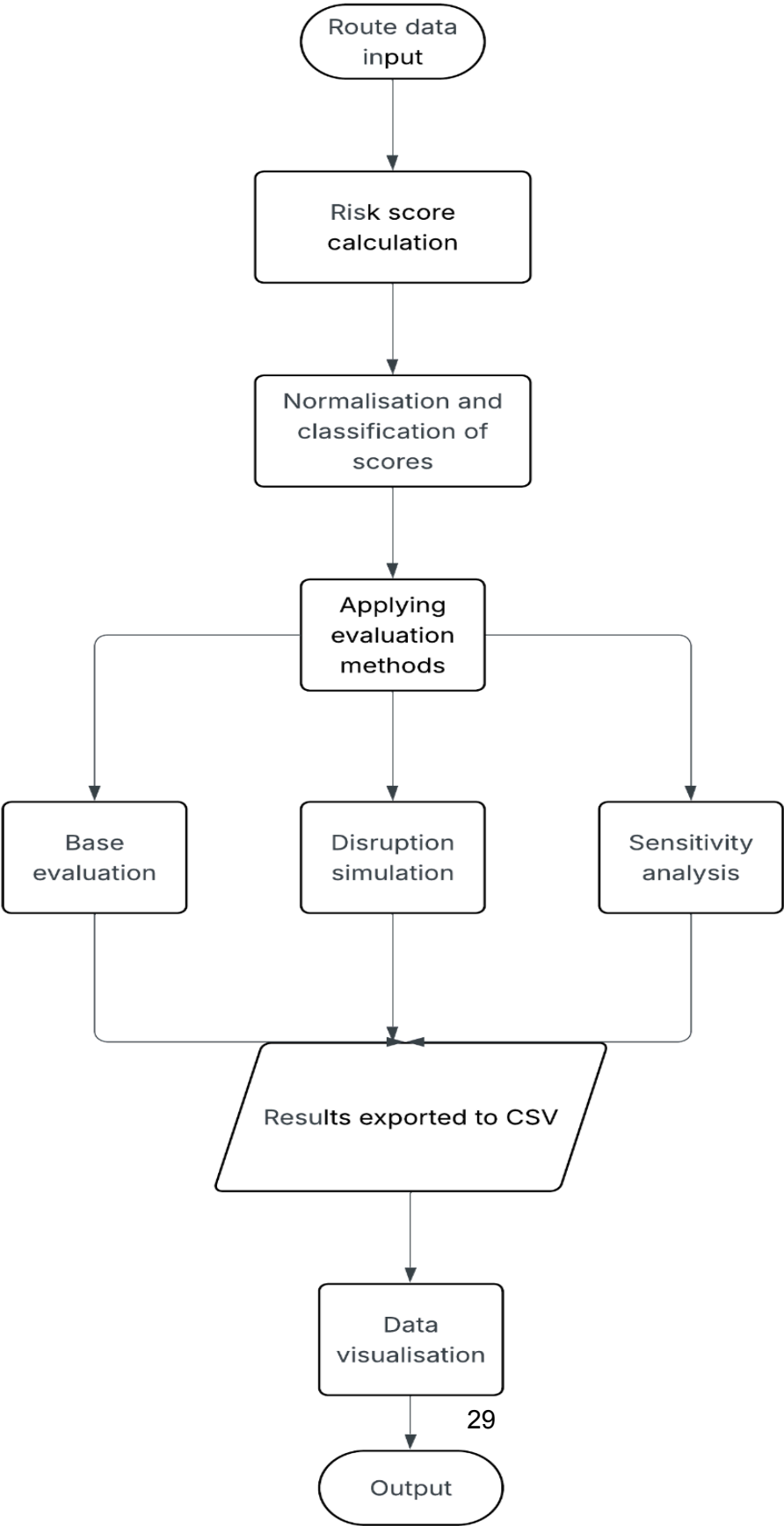


Figure 6: The functional components of the model presented in a pipeline visualisation

This flowchart demonstrates the steps taken to implement the proposed solution, corresponding to the objectives established in Chapter 1.

4.2 Data Representation (Routes & Journeys)

Each route is being represented as a Python dictionary containing all properties relevant to its risk evaluation - the route name, origin and destination, lines used, a list of interchange dictionaries, estimated travel time, step-free status of both the origin and destination, and a notes field for contextual information. Each interchange is itself a nested dictionary specifying the station name, complexity classification, and number of lifts required.

```
route1 = {
    "name": "J1 Route 1",
    "start": "Vauxhall",
    "end": "Mile End",
    "lines": ["Victoria", "District"],
    "interchanges": [
        {
            "station": "Victoria",
            "complexity": "high",
            "lifts_required": 7
        }
    ],
    "estimated_time_mins": 32,
    "start_step_free": True,
    "destination_step_free": False,
    "notes": "Google Maps marked as accessible but Mile End is not fully step-free"
}
```

Figure 7: Python dictionary representing Journey 1 Route 1, and showing the route data structure used throughout the model.

In total, 4 journeys are defined, with each comprising three routes between the same origin and the same destination. These journeys were selected carefully to produce a varied range of risk profiles, with all routes sourced directly from existing journey planning systems including Google Maps and TfL Go, ensuring

the evaluation is based on real-world recommendations rather than hypothetical scenarios. Table 2 summarises all twelve routes evaluated in this project.

Journey	Route	Time (mins)	Interchanges	Destination step-free
Vauxhall -> Mile End	Journey 1 Route 1	32	Victoria (High)	No
Vauxhall -> Mile End	Journey 1 Route 2	56	Green Park (Med), Canary Wharf (Med)	Yes (achieved by bus)
Vauxhall -> Mile End	Journey 1 Route 3	55	Bank (High), Westferry (Low)	Yes (achieved by bus)
Wembley Park -> Elephant & Castle	Journey 2 Route 1	45	Waterloo (High)	Yes
Wembley Park -> Elephant & Castle	Journey 2 Route 2	40	Baker Street (High)	No
Wembley Park -> Elephant & Castle	Journey 2 Route 3	62	King's Cross (High), Farringdon (Med)	Yes
Kingsbury -> Paddington	Journey 3 Route 1	60	Green Park (Med), Bond Street (Med)	Yes
Kingsbury -> Paddington	Journey 3 Route 2	35	Bank (High), Westferry (Low)	Yes
Kingsbury -> Paddington	Journey 3 Route 3	61	Kilburn (Low)	Yes
Canary Wharf -> King's Cross	Journey 4 Route 1	27	Farringdon (Med)	Yes
Canary Wharf -> King's Cross	Journey 4 Route 2	26	Bank (High)	Yes
Canary Wharf -> King's Cross	Journey 4 Route 3	36	Baker Street (High)	Yes

Table 2: Summary of all twelve routes evaluated across the four journeys, showing estimated travel time, interchange stations with complexity classifications, and destination step-free status.

4.3 Risk Calculation Process

One of the most important parts of the system is the risk calculation function, which takes a route and an optional weights dictionary as input and returns a raw risk score alongside a list of reasons explaining each penalty applied. The weights parameter allows the function to be called with modified values during sensitivity analysis without making any change to the base model.

```
def compute_risk(route, weights=None):
    if weights is None:
        weights = {
            "interchange_base": 1,
            "complexity_high": 4,
            "complexity_medium": 2,
            "complexity_low": 1,
            "lift_high": 5,
            "lift_moderate": 3,
            "lift_low": 1,
            "destination_penalty": 14
        }

    if not route.get("start_step_free", True):
        return 25, ["Start is not step-free – journey not feasible"]
```

Figure 8: The compute_risk() function showing the default penalty weights dictionary, illustrating how each risk factor is assigned a configurable numerical value

The function applies penalties in sequence via interchange count, per-interchange complexity, tiered lift dependency based on the total lift count across all interchanges, and finally the destination accessibility penalty where applicable. Each penalty applied is recorded alongside its value, making the reasoning behind every score transparent and auditable. The raw score is passed to the

normalisation function, which maps it to a 0–10 scale using a maximum raw score of 25, and subsequently to the classification function, which assigns a low, medium, or high band based on fixed thresholds as defined in section 3.5.3.

The correctness of this logic was verified through a set of manual tests, each comparing the function's output against a hand-calculated expected value for a representative set of routes. All ten checks passed consistently across runs, confirming that the implementation correctly reflects the model defined in Chapter 3.

The disruption simulation extends this process by creating a deep copy of a route, modifying the lift count at a specified interchange station according to the chosen severity level, and rerunning the risk calculation on the modified route. This allows before and after scores to be compared without altering the original route data. The sensitivity analysis similarly calls the risk calculation function repeatedly with modified weight dictionaries, varying each penalty by +/- 20% in isolation to assess its influence on the final classification.

4.4 Example Outputs

The system produces three categories of output. The base evaluation prints a structured breakdown of each route's risk score and contributing penalties, and exports results to `base_scores.csv`. The disruption simulation tests each route under minor and major fault scenarios at specified interchange stations, recording before and after scores and flagging classification changes or infeasibility. The sensitivity analysis varies each penalty weight by +/- 20% in isolation and records whether any route's classification changes. All three outputs are exported to CSV files which were used as the data source for the visualisations presented in Chapter 5.

Figure 9 shows the base evaluation output for Journey 2, illustrating the model's auditability - every point contributed to a score is explicitly reported, making the reasoning behind each classification transparent and verifiable.

```

=====
Journey 2: Wembley Park -> Elephant & Castle
=====

J2 Route 1
Time: 45 mins
Risk: 3.2/10 (LOW)
- 1 interchange(s) (+1)
- Waterloo is high complexity (+4)
- Moderate lift dependency (3 lifts) (+3)

J2 Route 2
Time: 40 mins
Risk: 8.8/10 (HIGH)
- 1 interchange(s) (+1)
- Baker Street is high complexity (+4)
- Moderate lift dependency (2 lifts) (+3)
- Destination is not step-free (+14)

J2 Route 3
Time: 62 mins
Risk: 5.2/10 (MEDIUM)
- 2 interchange(s) (+2)
- King's Cross St. Pancras is high complexity (+4)
- Farringdon is medium complexity (+2)
- High lift dependency (6 lifts) (+5)

--- Summary ---
J2 Route 1: 45 mins, risk 3.2/10
J2 Route 2: 40 mins, risk 8.8/10
J2 Route 3: 62 mins, risk 5.2/10

Fastest: J2 Route 2
Safest: J2 Route 1

```

Figure 9: Console output of the base evaluation for Journey 2, demonstrating all three risk classifications across three routes and the model's per-penalty score breakdown.

The output demonstrates the model's key design property - that every classification is fully explainable. The contrast between Journey 2 Route 1 scoring 3.2 (Low) despite passing through the high complexity Waterloo interchange, and Journey 2 Route 2 scoring 8.8 (High) via a comparatively simpler Baker Street interchange, illustrates how the destination penalty dominates when present. A user or developer can trace exactly why each route receives its classification, making the model's reasoning transparent and auditable. This transparency is

particularly important in an accessibility context, where the consequences of an incorrect recommendation can be significant for a disabled passenger.

5: Evaluation

5.1 Evaluation Methodology

This chapter is based on evaluating the risk-based model developed in Chapters 3 and 4 against real journey scenarios on the London Underground, structured around three components: a base evaluation of all twelve routes across four journeys, a disruption simulation testing route reliability under infrastructure fault scenarios, and a sensitivity analysis assessing the robustness of the model's penalty weights. Routes are sourced directly from Google Maps, Citymapper and TfL Go, ensuring the evaluation matches genuine recommendations currently being made to disabled passengers rather than artificially constructed scenarios. Journeys 1 and 2 have routes where the destination is not fully step-free, allowing the destination penalty to contribute to the score, while Journeys 3 and 4 are fully step-free end-to-end, isolating the interchange complexity and lift dependency factors.

Each journey was evaluated running the model against all three of its routes and comparing the resulting risk scores, classifications, and travel times. The disruption simulation was then applied to key interchange stations within each journey, testing both minor and major fault scenarios as defined in section 3.5.4. Finally, the sensitivity analysis varied each penalty weight by +/- 20% in isolation to assess whether the model's classifications are stable under changes to its assumptions.

The evaluation is primarily analytical than comparative - the goal is not to benchmark the model against existing journey planning systems, but to assess whether it produces meaningful, defensible, and consistent results that address the research question established in Chapter 1. Where findings are

counterintuitive or reveal limitations, these are discussed critically in section 5.5 and further addressed in Chapter 6.

5.2 Base Evaluation Results

Table 3 presents the risk evaluation results for all twelve routes across the four journeys. Figure 10 visualises the relationship between travel time and risk score across all routes.

Journey	Route	Time (mins)	Risk Score	Classification	Fastest	Safest
Journey 1 Route 1	Vauxhall → Mile End	32	9.6	High	✓	
Journey 1 Route 2	Vauxhall → Mile End	56	4.4	Medium		✓
Journey 1 Route 3	Vauxhall → Mile End	55	4.8	Medium		
Journey 2 Route 1	Wembley Park → Elephant & Castle	45	3.2	Low		✓
Journey 2 Route 2	Wembley Park → Elephant & Castle	40	8.8	High	✓	
Journey 2 Route 3	Wembley Park → Elephant & Castle	62	5.2	Medium		
Journey 3 Route 1	Kingsbury → Paddington	60	3.6	Low		
Journey 3 Route 2	Kingsbury → Paddington	35	3.2	Low	✓	
Journey 3 Route 3	Kingsbury → Paddington	61	1.2	Low		✓

Journey 4 Route 1	Canary Wharf → King's Cross	27	2.4	Low		✓
Journey 4 Route 2	Canary Wharf → King's Cross	26	3.2	Low	✓	
Journey 4 Route 3	Canary Wharf → King's Cross	36	3.2	Low		

Table 3: Risk evaluation results for all twelve routes across four journeys, showing normalised risk scores, risk classifications, and whether each route represents the fastest or safest option within its journey.

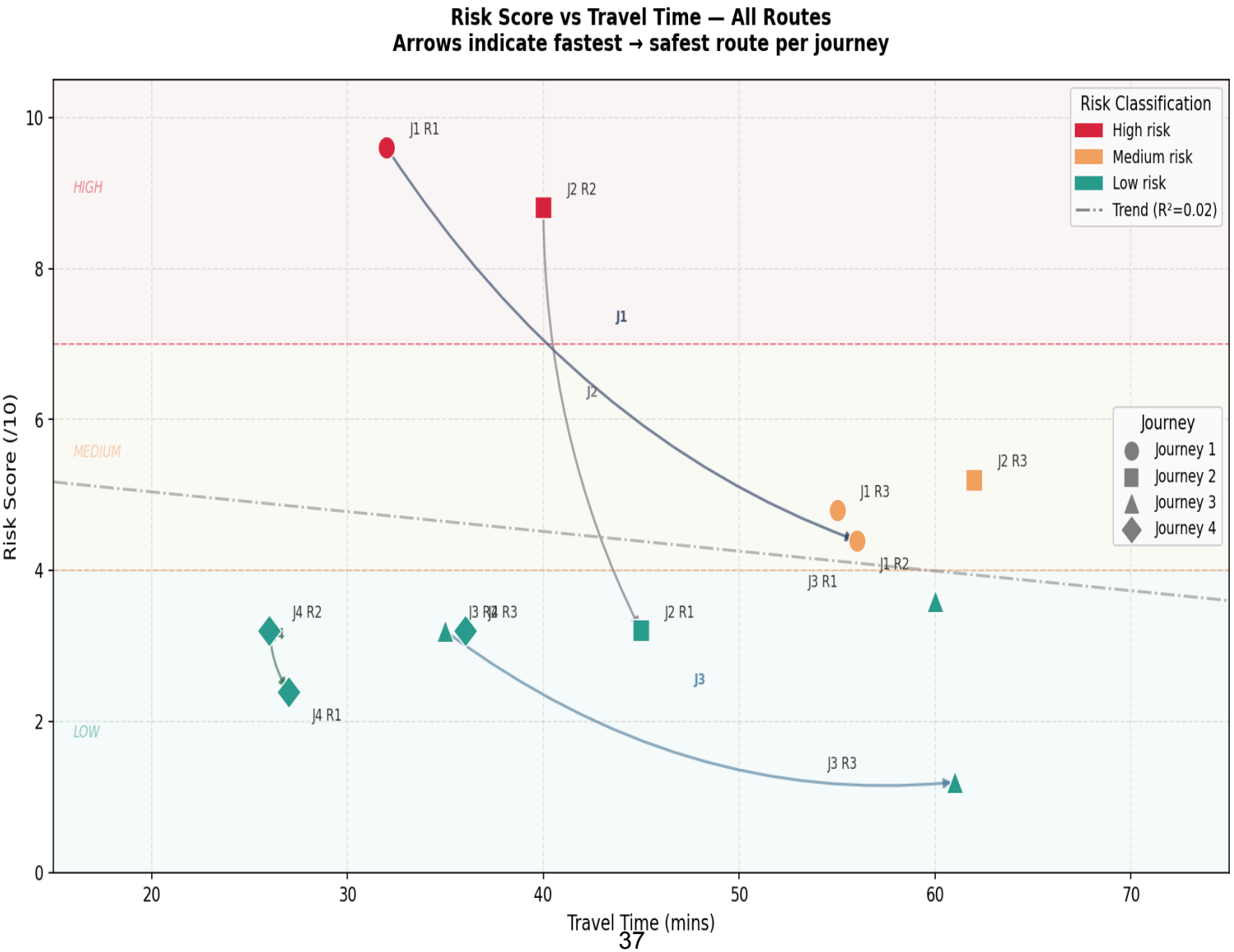


Figure 10: Scatter plot of risk score against travel time for all twelve evaluated routes. The trend line ($R^2=0.021$) indicates no meaningful correlation between journey duration and accessibility risk. Arrows indicate the fastest to safest route tradeoff within each journey.

The most immediate finding across all four journeys is that travel time and risk score show no meaningful correlation. A linear regression across all twelve routes produces an R^2 value of 0.021, indicating that travel time accounts for only 2.1% of the variance in risk scores. This is a significant finding - it demonstrates quantitatively that faster routes are not inherently riskier or safer, and that risk is determined by accessibility factors rather than journey duration. This directly challenges the implicit assumption in existing journey planning systems that the fastest accessible route is the most appropriate recommendation for a disabled passenger.

Journey 1 (Vauxhall → Mile End)

Journey 1 produces the widest risk spread of any journey evaluated, ranging from 4.4 to 9.6. Journey 1 Route 1 is the fastest at 32 minutes but scores 9.6/10 (High), driven primarily by the destination penalty - Mile End station lacks full step-free access to street level, a fact that Google Maps fails to communicate to users as demonstrated in section 2.5.3. This finding directly validates the model's core argument: a route that existing systems classify as wheelchair accessible is identified by the model as 'High' risk and practically unusable for a passenger requiring step-free exit. Journey 1 Routes 2 and 3 both score MEDIUM at 4.4 and 4.8 respectively, and both provide fully step-free access to the destination via bus. The near-identical scores despite different interchange profiles - Route 2 uses two medium complexity interchanges while Route 3 uses one high and one low, demonstrating that the model captures interchange risk in a nuanced way rather than simply penalising route complexity uniformly.

Journey 2 (Wembley Park → Elephant & Castle)

Journey 2 is the most analytically rich of the four journeys, producing all three risk classifications across its three routes. Journey 2 Route 2 is the fastest at 40 minutes yet scores 8.8/10 (High) - the second highest score in the entire dataset due to the combination of Baker Street's high complexity interchange and the destination being inaccessible from the Bakerloo line. This is the clearest example in the evaluation of a time-optimised journey planner directing a disabled passenger toward the most dangerous available route. Journey 2 Route 1 scores just 3.2/10 (Low) despite passing through Waterloo, the station with the highest recorded lift fault count on the entire network (451 faults, Transport for London, 2026b). This finding reveals an important nuance, which is that Waterloo's high fault frequency does not automatically make it the riskiest interchange in this journey, because the destination is step-free and the route has sufficient lift redundancy to absorb a minor fault. Journey 2 Route 3 scores 5.2/10 (Medium) and represents the safest fully accessible route, albeit at 62 minutes.

Journey 3 (Kingsbury → Paddington)

Journey 3 produces all 'Low' classifications, reflecting the fact that all three routes are fully step-free end-to-end and the destination penalty never applies. Without this dominant factor, the model differentiates on interchange complexity and lift dependency alone, producing scores ranging from 1.2 to 3.6. Notably, Journey 3 Route 2 via Baker Street scores 3.2/10 (Low) and is the fastest at 35 minutes, making this the first journey in the evaluation where the fastest route is also among the safest. Journey 3 Route 3 via Kilburn scores the lowest of any route in the entire dataset at 1.2/10, reflecting its single low complexity interchange and minimal lift dependency. However as demonstrated in section 5.3, this apparently safe route conceals a critical vulnerability under disruption conditions.

Journey 4 (Canary Wharf → King's Cross St. Pancras)

Journey 4 reinforces the all-low pattern observed in Journey 3, with scores ranging narrowly from 2.4 to 3.2. All three routes are fully step-free and the destination penalty does not apply. Journey 4 Route 1 via Farringdon is both the second fastest and the lowest risk at 2.4/10, while Journey 4 Route 2 via Bank is

marginally faster at 26 minutes but scores 3.2/10 - demonstrating that even within a LOW risk journey, the model identifies a meaningful reliability difference between routes. This is the clearest example in the evaluation where a reliability-focused approach would recommend a route that differs from the time-optimised recommendation, even when both routes are fully step-free.

5.3 Disruption Testing Results

The disruption simulation extends the base evaluation by testing how route reliability changes when lift infrastructure fails at key interchange stations. Figure 11 presents the before and after risk scores for all disruption scenarios evaluated, distinguishing between minor faults, major faults, and cases where a route becomes entirely infeasible.

Figure 11: Route Risk Score Before and After Disruption Simulation

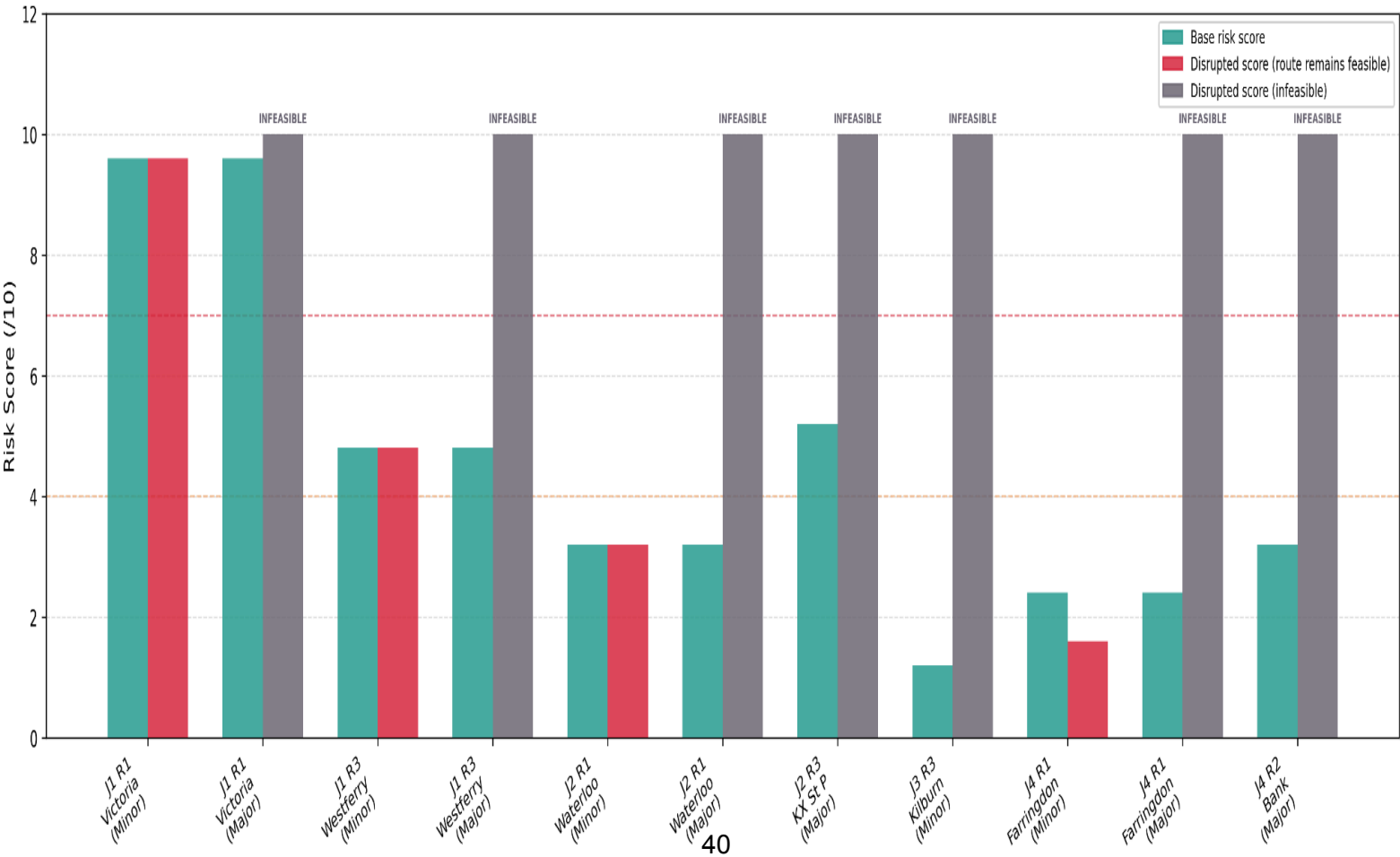


Figure 11: Route risk scores before and after disruption simulation across eleven scenarios, showing base score (teal), disrupted score where the route remains feasible (red), and disrupted score where the route becomes infeasible (grey).

The most significant finding from the disruption testing is that base risk scores alone are insufficient to assess real-world route reliability. This is most clearly demonstrated by Journey 3 Route 3, which scores 1.2/10 in the base evaluation - the lowest score in the entire dataset which becomes immediately infeasible under even a minor fault at Kilburn. Kilburn station has a single lift serving its single platform, meaning there is no redundancy whatsoever. A minor fault, defined in this model as a short-duration outage affecting one lift, is sufficient to render the entire route unusable for a disabled passenger. This finding surfaces a class of vulnerability that existing journey planning systems do not capture a route that appears maximally safe under normal conditions can collapse entirely under the most minor infrastructure failure.

The contrast with Journey 2 Route 1 via Waterloo is analytically important. Waterloo recorded the highest number of lift and escalator faults of any station on the network in 2025 (Transport for London, 2026b), yet Journey 2 Route 1 absorbs a minor fault at Waterloo without any change to its risk score or classification. This is because Waterloo's interchange involves three lifts, meaning the loss of one lift under a minor fault reduces the count from three to two, remaining within the same moderate lift dependency penalty tier. Only a major fault simulating an extended outage exceeding 20 hours renders Journey 2 Route 1 infeasible. This demonstrates that the model captures lift redundancy meaningfully, distinguishing between stations where a single fault is catastrophic and those where the infrastructure can absorb disruption without affecting journey feasibility.

The Victoria disruption scenario for Journey 1 Route 1 produces a stable result under a minor fault, with the score remaining at 9.6/10 (High). This is expected as the route is already classified 'High' primarily due to the destination penalty rather

than lift dependency, meaning a reduction in Victoria's lift count does not change the overall classification. Under a major fault however, Journey 1 Route 1 becomes infeasible, confirming that a route already identified as high risk is also highly susceptible to extended infrastructure outages.

The Farringdon scenarios across Journeys 2 and 4 produce contrasting results worth noting. In Journey 2, a minor fault at Farringdon leaves Journey 2 Route 3 stable at 5.2/10 (Medium), since Farringdon has two lifts and losing one keeps the route within the same penalty tier. In Journey 4, a minor fault at Farringdon on Journey 4 Route 1 reduces the score from 2.4 to 1.6 - the only scenario in the entire disruption evaluation where a minor fault causes a score reduction rather than an increase or infeasibility. This occurs because reducing Farringdon's lift count from two to one pushes it into the lower single-lift penalty tier, producing a counterintuitive result. This is acknowledged as a limitation of the tiered lift penalty approach and is discussed further in section 6.2.

Overall, the disruption testing confirms that the model's static base scores provide a useful but incomplete picture of route reliability. The simulation reveals that route vulnerability under disruption is determined not only by the number of lifts but by whether those lifts have redundancy - a distinction that the base scoring alone cannot fully capture.

5.4 Sensitivity Analysis

The sensitivity analysis evaluates the robustness of the model's penalty weights by varying each weight by $\pm 20\%$ in isolation and recording whether any route's risk classification changes as a result. The $\pm 20\%$ range was chosen to represent a reasonable bound on assumption error, and it is acknowledged that this threshold is not empirically derived, but reflects a meaningful degree of variation around the base weights given that no historical disruption data was available to calibrate them more precisely. Figure 12 presents the results for all twelve routes across all eight penalty variations.

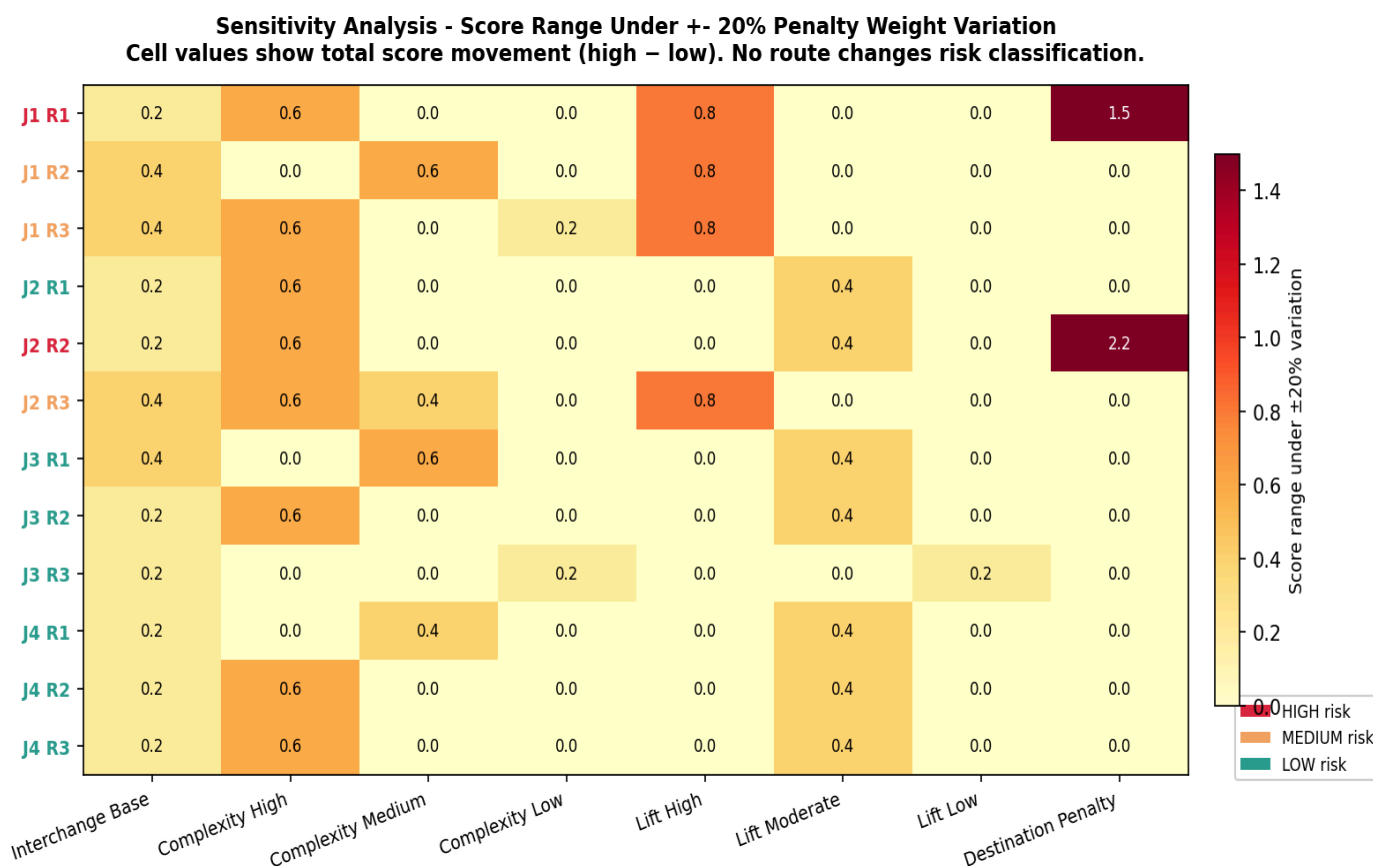


Figure 12: Heatmap showing total score movement under +- 20% penalty weight variation for all twelve routes and eight penalty weights. No route changes risk classification under any adjustment.

The key finding is that all twelve routes remain completely stable across every penalty variation tested - no route changes classification under any combination of weight adjustments within the +- 20% range. This is a strong indicator that the model's classifications are robust to reasonable variation in penalty assumptions. Considering that the penalty weights are assumption-based rather than empirically derived, as acknowledged in section 3.6.2, this stability provides meaningful evidence that the model's outputs are reliable within a reasonable range of weight variation.

However, this stability is not uniform across all risk bands, and the sensitivity analysis provides stronger evidence of robustness for High-risk classifications than for Low-risk ones. The complete stability across all routes is partly a consequence

of the destination penalty's dominance - routes that attract this penalty score so highly that moderate weight adjustments cannot move them across a classification boundary. For routes without the destination penalty, scores cluster in the Low band with limited spread, meaning weight adjustments produce only minor numerical changes without crossing thresholds. This introduces an important limitation of the sensitivity analysis itself: for Low-band routes, the test is less informative than it appears, because the absence of spread means classification stability reflects the structure of the scoring bands as much as genuine robustness of the weights. A more discriminating test for these routes would require either a wider variation range or simultaneous adjustment of multiple weights, neither of which was conducted here. The stability observed across the dataset as a whole should therefore be interpreted with this asymmetry in mind.

This finding is particularly significant for the 'High' risk classifications in the dataset. Journey 1 Route 1 scores 9.6/10 and Journey 2 Route 2 scores 8.8/10 - both remain firmly in the 'High' band regardless of weight adjustments, reflecting the dominant influence of the destination penalty in these routes. Even reducing the destination penalty by 20% to 11.2 points is insufficient to shift either route out of the High classification, confirming that these routes represent genuinely high risk under any reasonable parameterisation of the model.

This reinforces the finding from section 5.2 that the destination penalty is the most influential factor in the model. Future work could address this limitation by deriving weights empirically from historical TfL fault data, which would both narrow the uncertainty around the penalty values and make the sensitivity analysis more meaningful across the full range of risk classifications produced by the model.

5.5 Discussion of Findings

The evaluation presented in this chapter addresses the research question established in Chapter 1: **How can accessibility-related risk factors be modelled and integrated into journey planning systems in order to improve the reliability and usability of accessible routes for disabled users?** The

results demonstrate that such modelling is both technically feasible and analytically meaningful, producing outputs that existing journey planning systems are not capable of generating. Critically, all routes evaluated were sourced directly from Google Maps and TfL Go, meaning the findings represent a direct re-evaluation of recommendations currently being made to disabled passengers rather than a comparison against hypothetical scenarios.

The most significant finding across the entire evaluation is that the step-free status of the destination station is by far the most influential risk factor in the model. Routes where the destination lacks full step-free access to street level score between 8.8 and 9.6, consistently in the High band, while fully step-free routes score between 1.2 and 5.2 regardless of interchange complexity or lift dependency. Geurs and van Wee (2004) note that accessibility is frequently reduced to binary measurable indicators for evaluation purposes - the findings of this model demonstrate precisely why that reduction is problematic, producing risk scores that differentiate between routes that binary systems treat as equivalent. The model assigns a concrete, comparable score to this misrepresentation for the first time, demonstrating that what Google Maps classifies as a wheelchair accessible route is in practice a High risk journey that may be entirely unusable for a disabled passenger requiring step-free exit from the destination station, as illustrated in section 2.5.3.

The second key finding is that travel time and risk score are not correlated across the routes evaluated. The R^2 value of 0.021 confirms that journey duration accounts for only 2.1% of the variance in risk scores, providing quantitative evidence that the fastest accessible route is not reliably the safest. This directly challenges the design assumption underlying Google Maps, TfL Go, and Citymapper - all of which present routes ordered by travel time as identified in section 2.5.1. Since the routes evaluated were sourced directly from these systems, this finding demonstrates that the time-optimised recommendations currently being served to disabled passengers are in at least two of the four journeys evaluated the highest risk options available - a finding with direct practical implications for the 1.2 million disabled Londoners who identify as disabled under

the Equality Act 2010 and rely on these tools daily (ONS 2023, cited in Chapter 1).

The disruption simulation extends this finding further, revealing that base risk scores alone are insufficient to fully characterise route reliability. Journey 3 Route 3 demonstrates this most starkly - scoring 1.2 in the base evaluation yet becoming immediately infeasible under a minor fault at Kilburn. This surfaces a category of vulnerability that no existing journey planning system captures: routes that appear maximally safe under normal conditions but depend on a single point of infrastructure with no redundancy. Yet despite the well-documented frequency of lift faults on the London Underground network, existing journey planning systems continue to present accessible routes without any consideration of the disruptions a station may currently be experiencing or its historical susceptibility to infrastructure failure. This finding aligns with the argument made by Prakash, Mehta and Sharma (2024) that route feasibility in real-world transport systems is inherently dynamic, and that static accessibility classifications are insufficient to represent the true reliability of a journey. The model's disruption findings are particularly significant in light of Transport for London's own acknowledgement that when accessible routes fail entirely, staff must arrange alternative transport for affected passengers (Transport for London, 2026a) - implicitly confirming that route infeasibility under disruption is a known and recurring real-world phenomenon rather than a theoretical concern.

The sensitivity analysis confirms that the model's classifications are robust - complete stability across all twelve routes under $\pm 20\%$ weight variation provides evidence that the findings are not artefacts of arbitrary penalty choices. This is particularly important given the assumption-based nature of the weights, as it demonstrates that the model's core conclusions would hold under a reasonable range of alternative parameterisations.

It should be noted that the model evaluates routes as static entities - it does not account for individual passenger needs, time-of-day variation in fault probability, or the cumulative effect of multiple disruptions occurring simultaneously, all of which

would affect real-world journey reliability. These limitations are discussed further in section 6.2.

Taken together, these findings suggest that a reliability-focused approach to accessible journey planning is not only technically achievable but produces meaningfully different recommendations to those generated by existing systems when applied to routes those systems currently recommend. Specifically, this project demonstrates that three measurable route properties - interchange complexity, lift dependency, and destination accessibility - are sufficient to produce risk classifications that meaningfully diverge from time-optimised recommendations in at least two of the four journeys evaluated, suggesting that even a lightweight reliability scoring layer added to existing journey planning systems could meaningfully improve outcomes for disabled passengers. The model proposed in this project represents a proof of concept for integrating accessibility risk into route evaluation in a transparent, auditable, and reproducible way - directly addressing the gap identified in the literature review and providing a foundation for future development of journey planning tools that better serve the needs of disabled passengers.

6: Discussion and Conclusion

6.1 Summary of Findings

This project set out to investigate whether accessibility-related risk factors could be modelled and integrated into journey planning systems in a way that improves the reliability and usability of accessible routes for disabled users. The results demonstrate that this is both technically achievable and practically meaningful.

The base evaluation across four journeys and twelve routes, all sourced directly from existing journey planning systems, reveals that travel time and risk score are not correlated ($R^2=0.021$), and that in two of the four journeys the fastest route is also the highest risk. The model correctly identifies routes that Google Maps and TfL Go classify as wheelchair accessible but which are in practice unusable for passengers requiring step-free exit from the destination station. The disruption

simulation demonstrates that base risk scores alone are insufficient to characterise route reliability, with Journey 3 Route 3 scoring 1.2 in the base evaluation yet becoming immediately infeasible under a minor fault at Kilburn. The sensitivity analysis confirms that all twelve route classifications are completely stable under $\pm 20\%$ weight variation, providing evidence of model robustness despite the assumption-based nature of the penalty weights.

6.2 Limitations

Several limitations of the model have to be acknowledged. Penalty weights are based on literature and logical reasoning rather than empirically derived from historical disruption data, meaning the specific values assigned to each factor remain assumptions. Although the sensitivity analysis demonstrates stability under moderate variation, this does not eliminate the assumption-based nature of the weights entirely. Interchange complexity is assigned manually based on station layout documentation, annual passenger data, and lift fault frequency. This classification is informed by evidence but remains a judgment-based process rather than a formally computed metric, introducing subjectivity into a core component of the model.

Then, the model evaluates routes as static entities and does not account for time-of-day variation in fault probability, simultaneous disruptions across multiple interchanges, or individual passenger needs such as the specific type of mobility aid used. In practice, all of these factors would influence the reliability of an accessible journey. Other than that, the tiered lift penalty produced a counterintuitive result in one scenario - a minor fault at Farringdon on Journey 4 Route 1 reduces the score from 2.4 to 1.6 rather than increasing it, because removing one lift from a two-lift station pushes it into a lower penalty tier. This highlights a structural limitation of the tiered approach. A more consistent alternative would be a continuous penalty function such as $\text{penalty} = 1 + (1 / \text{lifts_required})$, where the penalty decreases monotonically as lift count increases, eliminating the possibility of a fault producing a lower score. This was not implemented within the scope of this project, as the focus was on demonstrating

the viability of the risk-based framework rather than optimising individual penalty functions, but represents a clear direction for future refinement.

Finally, the evaluation is limited to twelve manually defined routes across four journeys, introducing selection bias. The routes were chosen to demonstrate analytical variety rather than to represent a statistically comprehensive sample of all possible accessible journeys on the network.

Beyond technical limitations, any future deployment of this model raises legal, social, and ethical considerations worth acknowledging. If extended to incorporate real-time TfL API data, the system could log the journey behaviour of disabled passengers, introducing data protection obligations under GDPR that would need to be addressed before public use. More broadly, the findings of this model surface an equity concern: accessibility risk is not evenly distributed across the network, and the evidence that existing systems direct disabled passengers toward the highest risk routes in two of the four journeys evaluated suggests a systemic gap in how these tools are designed and regulated. As learnt from the Gebresselassie and Baljko paper, no regulatory obligation currently exists requiring third-party developers to prioritise accessibility in journey planning tools, meaning improvements depend entirely on developer discretion. Finally, improving the reliability of accessible public transport routes carries sustainability implications, as greater confidence in accessible journey planning reduces dependence on less sustainable door-to-door alternatives for disabled passengers.

6.3 Future Work

Several directions for future development follow naturally from the limitations identified above. The most impactful extension would be the integration of real-time lift status data via the TfL open data API, which publishes live lift availability across the network. This would transform the model from a static planning tool into a dynamic journey reliability companion, automatically applying disruption penalties to interchanges where lifts are currently out of service at the time of planning.

A second direction would be the empirical derivation of penalty weights from historical TfL fault data, correlating fault frequency and repair time at specific stations with their impact on journey usability. This would replace the current assumption-based weights with statistically grounded values, directly addressing the primary limitation of the model.

Third, graph-based route generation using an algorithm such as Dijkstra's, modified to filter non-step-free nodes, would allow the model to automatically enumerate all viable accessible paths between any origin and destination rather than relying on manually defined routes. This would eliminate selection bias and make the system scalable to the full London Underground network.

Finally, a web-based interface integrating the risk scoring model with the TfL API would allow disabled passengers to receive real-time, reliability-scored route recommendations directly, representing the practical realisation of the proof of concept demonstrated in this project.

6.4 Conclusion

This project has demonstrated that accessibility-related risk factors can be modelled in a transparent, reproducible, and analytically meaningful way. The risk-based evaluation model developed here produces outputs that differ meaningfully from the time-optimised recommendations of existing journey planning systems, identifying routes that are classified as accessible but are in practice 'High' risk or entirely infeasible under common disruption conditions. The disruption simulation and sensitivity analysis extend the model beyond static scoring, surfacing hidden vulnerabilities and confirming the robustness of the classifications produced.

The findings directly address the research question posed in Chapter 1, providing evidence that a lightweight reliability scoring layer, built on three measurable route properties is sufficient to produce more honest and practically useful route recommendations for disabled passengers. While the model represents a proof of

concept with acknowledged limitations, it establishes a clear and technically grounded foundation for future development of journey planning tools that prioritise reliability alongside efficiency, and that better reflect the real-world experience of the 1.2 million Londoners who identify as disabled under the Equality Act 2010 (ONS, 2023) - a population for whom unreliable accessibility information is not an inconvenience but a barrier to independent travel, and one that transport planning tools have a responsibility to serve more honestly.

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Appendix

Testing function

```
def run_manual_testing(route, expected_raw, expected_normalised, expected_class):
    raw, _ = compute_risk(route)
    normalised = normalize_risk(raw)
    classification = classify_risk(normalised)

    raw_ok = raw == expected_raw
    norm_ok = normalised == expected_normalised
    class_ok = classification == expected_class

    print(f"\n Test Case: {route['name']}")
    print(f"Raw:          {raw} (expected {expected_raw}) {'✓' if raw_ok else 'X'}")
    print(f"Normalised:      {normalised} (expected {expected_normalised}) {'✓' if norm_ok else 'X'}")
    print(f"Class:           {classification} (expected {expected_class}) {'✓' if class_ok else 'X'}")
```

Each test compares the function's computed raw score, normalised score, and classification against a hand-calculated expected value.

Results

Test	Route	Expected Raw	Expected Normalised	Expected Classification	Result
1	J1 Route 1	24	9.6	High	✓
2	J1 Route 2	11	4.4	Medium	✓
3	J1 Route 3	12	4.8	Medium	✓
4	J2 Route 2	22	8.8	High	✓
5	J3 Route 3	3	1.2	Low	✓
6	J3 Route 2	8	3.2	Low	✓
7	J3 Route 1	9	3.6	Low	✓
8	J4 Route 1	6	2.4	Low	✓
9	J4 Route 2	8	3.2	Low	✓
10	J4 Route 3	8	3.2	Low	✓

ECS635U – Project Module
Generative AI Tools – Student Accountability Statement

The completed and signed statement must be included in the appendix of the progress and final reports.

Student Name: Mateusz Artur Wasielewski

Project Title: Beyond Step-Free: A risk-based framework for evaluating accessible journey reliability on the London Underground

Supervisor Name: Tayyaba Irum

1. Have you used Generative AI tools for your project?

YES ✓

NO

If yes, please complete sections 2 and 3 and sign the declaration below.

If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
ChatGPT	3.5	OpenAI	https://chat.openai.com/
Semantic Scholar	N/A	Allen Institute for AI	https://www.semanticscholar.org

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- 1 Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- 2 Software development (e.g. AI-assisted code generation, debugging, documentation)
- 3 Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- 4 Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- 5 The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Project Section	Description	Justification
Methodology	I have used AI in order to check for grammar and spelling, and added certain phrasing suggestions made by OpenAI	This was used solely to improve clarity and for me to ensure the project, especially at its most important stage, is as cohesive as possible; all content and arguments are my own.
Literature Review	I have used an AI tool that searches and finds full, downloadable PDF versions of existing research papers (Semantic Scholar)	This was used to locate and access full PDF versions of research papers after I realised a big proportion of existing research made on this topic is hardly accessible from websites such as ScienceDirect. This saved time in sourcing literature but all reading, and analysis of the papers is my own.
Code	I have used OpenAI for help with certain syntax errors encountered while implementing	This was used only to assist with occasional debugging, whereas all logic, structure, and design decisions are my own original work.

3. Generative AI Student Declaration

- 1 I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- 2 I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- 3 I take full responsibility for the originality and integrity of my project submission.
- 4 I have referenced all sources, including those suggested or produced with AI support.
- 5 I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- 6 My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: Mateusz Wasielewski Date: 06/05/2026

